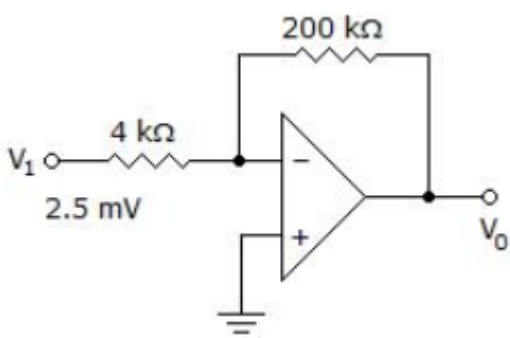




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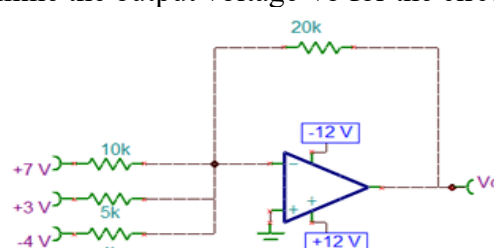
Date : 19/05/2025	CIE- 2	Maximum Marks : 60
Semester : 2	UG	Duration : 2 Hrs
Course :	Principles of Electronics Engineering	Course Code :EC123ATC

SL. NO	Part-A Quiz	M	BT	CO
1.	If two points V1 and V2 to an Op-Amp are 1.999 V and 2.001 V respectively. If amplifier has differential gain of 1000, then output voltage is ____ V.	1	3	1
2.	The output voltage of circuit given in Fig.1 with a sinusoidal input of 2.5 mV in Volts is ____.	1	4	2
	 Fig.1			
3.	The output impedance of an Ideal op-amp = ____.	1	2	1
4.	With a differential gain of 50,000 and a common-mode gain of 2, what is the common-mode rejection ratio in db?	1	3	4
5.	If an Op-Amp with an open loop gain of 99 is used as a voltage follower, then gain error ____ %.	1	4	3
6.	_____ provides physical connection between the transmitter and Receiver.	1	2	2
7.	The circuit that recovers the original modulating information from an AM signal is known as ____.	1	1	2
8.	_____ is a device used to convert the physical parameters in to an electrical signal.	1	1	3
9.	The register used for arithmetic and logical operations is called ____.	1	1	4
10.	The address of the next instruction is stored in the _____ register.	1	1	4

SL. NO	Part-B Test	M	BT	CO
1.a	Define the following terminology with respect to op-amp with necessary equations: i) Slew rate ii) CMRR The output signal of an op-amp with a slew rate of 2.5V/μs has a peak to peak value of 18V. Find the maximum frequency for undistorted output voltage.	6	2	1



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1.b	Determine the output voltage V_o for the circuit shown in Fig.2	4	4	2
 Fig. 2				
2.a	Draw the block diagram of a super heterodyne AM receiver and explain the function of each block.	6	2	1
2.b	Draw the circuit of an integrator using an op-amp and derive the expression for the output voltage.	4	3	2
3.a	Draw the general block diagram of a Communication System and explain the function of each stage.	6	1	4
3.b	A carrier of 1MHz with 400W of its power is amplitude modulated with a sinusoidal signal of 2500Hz. The depth of modulation is 75%. Calculate the sideband frequencies, the bandwidth, the power in the sidebands & the total power in the modulated wave.	4	3	4
4.a	Draw the circuit and calculate the values of different resistors of a summer circuit using two ideal op-amps to get $V_o = 5V_1 - 7V_2 + 9V_3 - 4V_4$.	6	4	4
4.b	Distinguish between AM and FM.	4	1	3
5.a	What is modulation? Mention its types and discuss why modulation is necessary in communication systems.	6	2	3
5.b	Difference between RISC and CISC architecture.	4	1	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

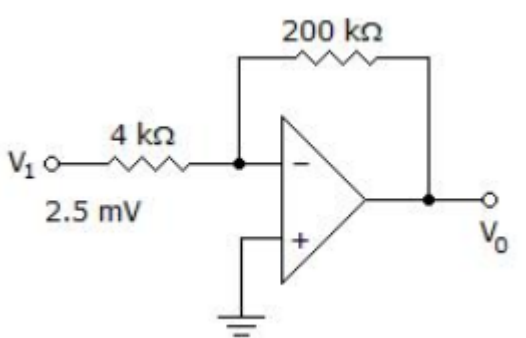
Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test-1	Max Marks-50	14	15	12	19	18	20	10	12	-	-

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SL. NO	Part-A Quiz	M	BT	CO
1.	If two points V1 and V2 to an Op-Amp are 1.999 V and 2.001 V respectively. If amplifier has differential gain of 1000, then output voltage is V. 2	1	3	1
2.	The output voltage of circuit given in Fig.1 with a sinusoidal input of 2.5 mV in Volts is _____. -0.125  Fig.1	1	4	2
3.	The output impedance of an Ideal op-amp = _____. 0	1	2	1
4.	With a differential gain of 50,000 and a common-mode gain of 2, what is the common-mode rejection ratio in db? 87.9	1	3	4
5.	If an Op-Amp with an open loop gain of 99 is used as a voltage follower, then gain error _____. 1	1	4	3
6.	_____ provides physical connection between the transmitter and Receiver. Communication Channel	1	2	2
7.	The circuit that recovers the original modulating information from an AM signal is known as _____. Demodulator	1	1	2
8.	_____ is a device used to convert the physical parameters in to an electrical signal. Input transducer	1	1	3
9.	The register used for arithmetic and logical operations is called _____. Accumulator	1	1	4
10.	The address of the next instruction is stored in the _____ register. Program Counter	1	1	4



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Sl. No.	Solution with Mark split-up	Max marks
1.a	i. Slew rate(S): It is defined as “The rate of change of output voltage per unit time” Ideally slew rate should be as high as possible. But its typical value is 0.5 V/μ-sec. -2M ii. Common Mode Rejection Ratio (CMRR): It is defined as “The ratio of differential voltage gain to common-mode voltage gain”. $\text{CMRR} = \text{Differential mode gain} / \text{Common-mode gain}$ $\text{CMRR in dB} = 20\log A_d/A_c$ -2M iii. Frequency=44.2KHz -2M	06
1.b	$V_O = -(R_f/R_1)(V_1) + R_f/R_2(V_2) + R_f/R_3(V_3) = -[14+12-20] = -6V$	04
2.a	Block diagram: 2M and Explanation:4M Superhetrodyne Receiver: a. In a super hetrodyne receiver, the incoming or selected radio frequency signal is mixed with a high frequency signal generated by a local oscillator present in the receiver itself. b. It can be seen that the variable capacitors of RF amplifier, mixer & local oscillator are all ganged to a common shaft so that the rotation of ganged shaft will simultaneously vary the capacitances of tank circuits of all 3 stages. c. The function of each block is as follows: i) RF amplifier stage: It functions like filter and amplifier. The receiving antenna picks the weak radio signals transmitted by various stages & feeds them to RF stage. The RF stage selects the desired radio signal using its LC tuning circuit & amplifier it to desired level. ii) Mixer Stage : This stage involves the mixing of RF input & local oscillator i/p. The 2 frequencies beat together & produce a constant intermediate frequency which is 455 KHz. i.e. $f_{IF} = f_{LO} - f_c = 455 \text{ KHz.}$ Where f_{IF} = intermediate frequency f_{LO} = Local oscillator frequency f_c = selected radio frequency. for eg: If it is desired to tune a station at 1MHz, RF stage LC is turned to resonate at this frequency. Now local oscillator is tuned to 1455 KHz & thus the mixer stage will o/p a frequency of 455KHz. iii) IF amplifier Stage: The resulting 455 KHz Signal is amplified by the IF amplifier & fed to the detector. iv) Detector: The function of detector is demodulation. The function of this stage is to recover AF signal or modulating signal that contains information from o/p signal of IF stage. v) AF amplifier stage : AF signal o/p of detector stage is amplified by this stage to a level sufficient to drive the speaker.	06



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	The elements of communication system are as follows: Input transducer: Is a device used to convert the physical parameters in to an electrical signal. Transmitter: It processes the input signal in a form suitable for transmission over the communication channel. This operation is termed as modulation. Communication channel: It provides physical connection between the transmitter and receiver. There are 2 types of channels: Point to point channel: Telephone wires, cables. Broad cast channel: Satellite in geostationary orbit. Noise: Is an unwanted energy that gets added to the message signal during transmission over the channel It is random in nature & has its greatest effect when the message signal is weakest. Receiver: It processes the signal & produces an estimate of original message signal This operation is called demodulation.	
3.b	Soln: $f_{USB} = f_c + f_m = 1002.5\text{KHz} \quad -0.5\text{M-}; \quad f_{LSB} = f_c - f_m = 997.5\text{KHz} \quad -0.5\text{M-};$ $BW = 2f_m = 5\text{KHz} \quad -1\text{M-};$ $P_t = 512.5\text{W} \quad -1\text{M-};$ $PSB = P_t - P_c = 112.5\text{W} \quad -1\text{M-}$	4
4.a	Soln: $V_o = 5V_1 - 7V_2 + 9V_3 - 4V_4.$ $R_f / R_1 = 5, R_f / R_2 = 7, R_f / R_3 = 9, R_f / R_4 = 4 \quad \text{-----}1\text{M}$ Choose $R_f = 10\text{K}\Omega, R_1 = 2\text{K}\Omega, R_2 = 1.43\text{K}\Omega, R_3 = 1.12\text{K}\Omega, R_4 = 2.5\text{K}\Omega, \quad \text{---}2\text{M}$ Circuit diagram----3M	6



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4.b	<table> <tr> <th>AM</th><th>FM</th></tr> <tr> <td>Amplitude of the carrier varies while its frequency remains constant</td><td>Frequency of carrier changes while its amplitude remains constant.</td></tr> <tr> <td>Modulation index can have values from 0 to 1 only</td><td>Modulation index is much greater than 1</td></tr> <tr> <td>AM has only 2 side bands</td><td>FM has infinite number of side bands</td></tr> <tr> <td>BW is twice the highest modulating frequency. $BW = 2f_m$</td><td>BW is much larger than AM $BW = 2(f_m + f_c)$</td></tr> <tr> <td>Carrier frequency is lower RF 500KHz to 3MHz</td><td>Carrier frequency is very high VHF & UHF(88 to 100MHz)</td></tr> <tr> <td>Susceptible to noise</td><td>More immune to noise</td></tr> <tr> <td>Propagation is by ground waves and sky waves</td><td>Propagation is by space waves</td></tr> <tr> <td>Requires less complex & less expensive equipment</td><td>Requires more complex and expensive equipment.</td></tr> </table>	AM	FM	Amplitude of the carrier varies while its frequency remains constant	Frequency of carrier changes while its amplitude remains constant.	Modulation index can have values from 0 to 1 only	Modulation index is much greater than 1	AM has only 2 side bands	FM has infinite number of side bands	BW is twice the highest modulating frequency. $BW = 2f_m$	BW is much larger than AM $BW = 2(f_m + f_c)$	Carrier frequency is lower RF 500KHz to 3MHz	Carrier frequency is very high VHF & UHF(88 to 100MHz)	Susceptible to noise	More immune to noise	Propagation is by ground waves and sky waves	Propagation is by space waves	Requires less complex & less expensive equipment	Requires more complex and expensive equipment.	4
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Propagation is by ground waves and sky waves	Propagation is by space waves																			
Requires less complex & less expensive equipment	Requires more complex and expensive equipment.																			
5.a	Soln: 1M+1M+4M Sol: Modulation: It is defined as the process by which some characteristic or property of a high frequency sine wave called the carrier is varied in accordance with instantaneous amplitude of the message signal called the modulating signal. - The characteristics of the carrier wave that is modified may be amplitude, frequency or phase angle. - Three types of modulation: 1. Amplitude modulation 2. Frequency modulation 3. phase modulation Sol: Direct transmission of information signals such as voice, video or binary data would result in catastrophic interference problems when transmission distances are large. - To carry out reliable long - distance radio communication. It is necessary to modulate the signal because of following reasons. To reduce the height of antenna: - For efficient radiation & reception the antenna's should have heights comparable to quarter wave length of the frequency used Minimum height of antenna = $\lambda/4 = c/4f$ - Where λ is the wavelength, f is the frequency of the signal & c is velocity of electro magnetic wave (light). To avoid mixing of Signals. - We know that frequency range of audio signal is from 20 Hz to 20 KHz. If there are several stations operating over this same frequency range, the programme of different stations will get mixed up. To increase the range of communication: - At low frequencies radiation is poor & signal. gets highly attenuates. Modulation makes it easier for Signals to be transmitted over long distance. To allow multiplexing of signals: - Multiplexing means transmission of two or more signals simultaneously over the same channel. Ex: Radio station broad Casting the signal in Mw & Sw band simultaneously. To allow adjustments in the BW (Band width): - Bw of a modulating signal may be made smaller or larger than the original signal. Improve quality of reception: - Reduction in noise improve the quality of reception.	6																		



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5.b	RISC	CISC	4
	Instruction takes one or two cycles	Instruction takes multiple cycles	
	Only load/store instructions are used to access memory	In addition to load and store instructions, memory access is possible with other instructions also	
	Instructions executed by hardware	Instructions executed by the microprogram	
	Fixed format instruction	Variable format instructions	
	Few addressing modes	Many addressing modes	
	Few instructions	Complex instruction set	
	Most of them have multiple register banks	Single register bank	
	Highly pipelined	Less pipelined	
	Complexity is in the compiler	Complexity in the microprogram	



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SCRUTINY & EVALUATION OF CIE QUESTION PAPER**To be filled by the Course handling faculties:**

As the faculty members of the course **Principles of Electronics Engineering** with code: **EC123ATC** we hereby confirm that the question paper is thoroughly reviewed and we ensured that it adheres to the following criteria:

S. NO	Criteria	Yes/ NO
1	The question paper adequately covers the prescribed syllabus contents.	Yes
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	Yes
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analyzing, evaluating, and creating.	Yes
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	Yes
5	Course handling faculty (As applicable) are responsible for preparing the question paper, scheme, and solution have unanimously agreed to utilize this question paper for conducting the Continuous Internal Evaluation.	Yes
6	The question paper, Scheme and Solution has been submitted to the Test coordinators within the designated timeframe to ensure the smooth conduction of Continuous Internal	Yes

Course handling Faculties

1. Dr. Shylashree N
2. Dr. Geetha K S and Prof. Sindhu Rajendran
3. Dr. Rajasree P M

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To be filled by the Scrutinizer's:

S. NO	Rubrics	Max points	Awarded
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams	10	
5	Scheme & Solution in the format	10	
	Total points	50	
Any other comments by the scrutinizers :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solution.

All the suggested corrections are incorporated and timely submitted	Yes/ NO	Accepted/ Rejected
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Signature of Scrutinizer**(Name:)****Signature of HOD**